

maximize the loss fⁿ
w.r.t D.

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim P(\text{data})} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log (1 - D(G(z)))]$$

minimize
the loss fⁿ
w.r.t. G

Overall
loss fⁿ

depends on
both D & G

E_x - expectation over real data distribution

E_z - " " random noise distribⁿ

$\log D(x)$ whr x is real data.

discriminator checks real data

0 = fake 1 = real

① if correctly op's real → loss is small
[good for D]

② if mistakenly op's fake → score is low.
loss is high
[bad for D]

What D wants? $\Rightarrow$ low loss.

$$\log(1 - D(G(z)))$$

Represents generator's loss.

1 - Discriminator's o/p.

(1) G creates fake images & sends it to D.

(2) D gives 0 for fake 1 for real.

(3) if D o/p's correctly says fake (0)
then loss is high
[Bad for G]

(4) if D mistakenly says real (1) $\rightarrow$
loss is low
[good for G]

What G wants? $\Rightarrow$ low loss.

so, eqⁿ sets up min max game betⁿ G & D.

⇒ G tries to minimize 2nd term (its loss) by producing increasingly realistic data that w^l confuse D.

⇒ D tries to maximize overall loss which includes its own loss in 1st term by improving its ability to identify fake data.

Numerical 1

Given : $D(x) = 0.9$ D's o/p on real ing
 $D(G(z)) = 0.3$ — " — fake ing
 calculator D's loss.

Step 1 : calculate D's loss

$$\text{Loss}_D = - [\log D(x) + \log (1 - D(G(z)))]$$

$$= - [\log (0.9) + \log (1 - 0.3)]$$

$$= - [\log (0.9) + \log (0.7)]$$

$$= - (-0.105 + (-0.357))$$

$$= 0.462$$

D's loss

0/1 for fake data
(should be close to 0)

$$\text{Loss}_D = - [\ln(D(x)) + \ln(1 - D(G(z)))]$$

↑
0/1 when
real data is
given (should
be close to 1)

D tries to

→ ↑ $\ln(D(x))$ correct for
real.
→ ↓ $\ln(1 - D(G(z)))$ correct
for fake

G' loss

$$\text{Loss}_G = -\ln(D(G(z)))$$

maximize $D(G(z))$

why? $\rightarrow$ G wants D to believe the fake is real.

so G tries to minimize

$$-\ln(D(G(z)))$$

In original papers $G_{\text{Loss}} = \ln(1 - D(G(z)))$
 because vanishing gradients so commonly used is $-\ln(D(G(z)))$.